PROBLEM 1.1. A simple neural network

a) Use `vmap` to compute an element-wise arctan activation function over a 1-D array.

$$\sigma(t) = \arctan(x) \tag{1}$$

b) Use `grad` to compute the gradient of the arctan function and verify with a plot that it matches the analytical solution given by

$$\sigma'(t) = \frac{1}{1+t^2}. \tag{2}$$

c) Write a `jit`-compiled dense layer function with $ReLU$ activation. Pass the weights and biases as a single array.

Hint: Do not use an activation function for the last layer, as it prohibits negative output values, which occur in the training data.

d) Concatenate 3 Layers, initialise the weights randomly and optimise them by writing a gradient descend loop over a least squares loss function. Make use of `jax.grad`. The data for fitting is provided. Visualise convergence by plotting the loss over the iteration number.

e) Redo Task d), but use the ADAM optimiser provided by the `optax` library[1].

f) Work through the TUWEL course *dataLAB Cluster Essentials*[2]. Perform the steps in section "Accessing the cluster" except of writing in the matrix channel. This will be done collectively by the supervisors once everybody finished the intro course.

[1] https://optax.readthedocs.io/en/latest/index.html

[2] https://tuwel.tuwien.ac.at/course/view.php?id=74282 Enrollment key: dataLABW2025